Unit - 6
Gaseous State

Evaluate yourself:

1. Volume of freon (V_1) = 1.5 dm^3

   Pressure (P_1) = 0.3 atm

   'T' isconstant

   P_2 = 1.2 atm

   V_2 = ?

   $\therefore P_1V_1 = P_2V_2$

   $$\Rightarrow V_2 = \frac{P_1V_1}{P_2}$$

   $$= \frac{0.3\ \cancel{atm} \times 1.5\ dm^3}{1.2\ \cancel{atm}}$$

   $$= 0.375\ dm^3$$

   $\therefore$ Volume decreased from 1.5 dm^3 to 0.375 dm^3

2. V_1 = 0.375 dm^3 V_2 = 0.125

   P_1 = 1.05 atm P_2 = ?

   'T' - Constant

   $P_1V_1 = P_2V_2$

   $$\therefore P_2 = \frac{P_1V_1}{V_2} = \frac{1.05 \times 0.375}{0.125}$$

   $$= 3.15\ atm$$

3. V_1 = 3.8 dm^3 $T_2 = 0^0C = 273$ K

   T_1 = ? V_2 = 2.27 dm^3

   $$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad T_1 = \left(\frac{T_2}{V_2}\right) \times V_1$$

   $$= \frac{273\ K}{2.27\ \cancel{dm^3}} \times 3.8\ \cancel{dm^3}$$

   $$T_1 = 457\ K$$

4. V_1 = 7.05 dm^3 V_2 = 2.35 dm^3

   η_1 = 0.312 mol η_2 = ?

   'P' and 'T' are constant

   $$\therefore \frac{V_1}{n_1} = \frac{V_2}{n_2}$$

   $$\Rightarrow n_2 = \left(\frac{n_1}{V_1}\right) \times V_2$$

   $$n_2 = \frac{0.312\ mol}{7.05\ \cancel{dm^3}} \times 2.35\ \cancel{dm^3}$$

   $$n_2 = 0.104\ mol$$

   Number of moles exhaled =

   0.312 - 0.104 = 0.208 moles

5) $T_1 = 8\ ^0C = 8 + 273 = 281$ K

   P_1 = 6.4 atm V_1 = 2.1 ml

   $T_2 = 25^0$ C = 25 + 273 = 298 K

   P_2 = 1 atm V_2 = ?

   $$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

   $$\Rightarrow V_2 = \left(\frac{P_1V_1}{T_1}\right) \times \frac{T_2}{P_2}$$

   $$= \frac{6.4\ \cancel{atm} \times 2.1\ ml}{281\ \cancel{K}} \times \frac{298\ \cancel{K}}{1\ \cancel{atm}}$$

   $$V_2 = 14.25\ ml$$

6(a)

Vo_2 = 12 dm^3 P = 1 atm

V_{He} = 46 dm^3

V_{total} = 5 dm^3

$$P_{O_2} = x_{O_2} \times P_{total} \qquad n_{O_2} = \frac{1\ mol}{22.4\ \cancel{L}} \times 12\ \cancel{L}$$

$$x_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{He}} \qquad n_{O_2} = 0.54\ mol$$

$$= \frac{0.54}{0.54 + 2.05} \qquad n_{H_e} = \frac{1\ mol}{22.4\ \cancel{L}} \times 46\ \cancel{L}$$

$$= \frac{0.54}{2.59} = 0.21 \qquad n_{H_e} = 2.05\ mol$$

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